

Supporting Information

Decoding the Biomimetic Mineralization of Metal-Organic Frameworks in Water

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1. Methodology Approach Section

1.1 Chemicals

All chemicals were obtained commercially and used without additional purification. 2-methylimidazole (HmIm, 99%) was purchased from Acros Organics. Hexa-decyltrimethylammonium bromide (CTAB, 98%) was purchased from Alfa Aesar. Zinc nitrate hexahydrate (99.0%) was purchased from SHOWA.

1.2 Synthesis of ZIF-8

All solutions were prepared on the day of the scattering SWAXS. Through our scattering experiments, we found that HmIm solutions stored for more than 2 to 3 days resulted in delayed overall reaction times. After sample preparation, they were stored at room temperature (28 °C) and in a refrigerator (10 °C). A solution cell with Kapton film was used for injecting samples. Experiments at 28 °C, 25 °C, 22.5 °C, and 20 °C were conducted using samples and a solution cell both equilibrated at 28 °C. While for the 10 °C, 12.5 °C, 15 °C, and 17.5 °C experiments, samples and a solution cell both equilibrated at 10 °C. A pre-equilibration period of 20 minutes was allowed to experiment target temperature before commencing the experiments. Throughout the process, a temperature control system equipped with both a chiller and a heater were used to maintain the samples and solution cell at the desired temperatures. Before injection into the solution cell, we rapidly prepared a homogeneous mixture comprising ZIF-8 samples with a Zn:HmIm molar ratio of 1:57 were synthesized by adding 0.25 mL of a 97.5 mM $\text{Zn}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ solution to a mixture of 1 mL of 0.9625 mM CTAB and 0.75 mL of 1.843 M 2-methylimidazole. This mixture was swiftly injected 100 μL into the solution cell to enable real-time observations. In-situ probing was conducted for 60 minutes during the synthesis of ZIF-8 at temperatures ranging from 10 °C to 22.5 °C and 30 minutes at 25 °C and 28 °C (room temperature).

1.3 Synthesis of ZIF-8 with pepsin and lysozyme

We prepared a homogeneous mixture comprising ZIF-8 samples with a Zn:HmIm molar ratio of 1:57 were synthesized by adding 0.25 mL of a 97.5 mM $\text{Zn}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ solution to a mixture of 1 mL of 0.9625 mM CTAB and 0.75 mL of 1.843 M 2-methylimidazole and 2 mg pepsin or lysozyme.

1.4 Synchrotron small- to wide-angle X-ray scattering (SWAXS) data collection

SWAXS analysis with synchrotron light source was applied to observe the metal-organic framework (MOF) nucleation. The data were collected at the TPS 13A BioSAXS beamline of the National Synchrotron Radiation Research Center (NSRRC), Taiwan. With an X-ray beam energy of 15.0 keV (wavelength $\lambda = 0.8266 \text{ \AA}$) and two sample-to-detector distances of 2713 and 286 mm with the Eiger X9M SAXS detector and the Eiger X1M WAXS detector, respectively, the merged data covered a wide scattering vector q -range of 0.007 to 2.0 $\text{\AA}^{-1}$. Another two sample-to-detector distances of 5993 and 643 mm with the Eiger X9M SAXS detector and the Eiger X1M WAXS detector, respectively, the merged data covered a wide scattering vector q -range of 0.0025 to 1.2 $\text{\AA}^{-1}$. The reduction of two-dimensional scattering data was processed to an incident count, which was corrected for background radiation, detector sensitivity, buffer transmission, buffer scattering, and sample trans-mission, and radially averaged to produce $I(q)$ vs. q profiles by using conventional procedures. In our experimental setup, simultaneous acquisition of SAXS and WAXS data was performed. Following the individual reduction of these scattering profiles, the SAXS measurements were analyzed preceded that of WAXS. To integrate the SAXS and WAXS observations effectively, we carefully aligned the overlapping q -range segments. This alignment was achieved through the meticulous adjustment of scaling factors, ensuring sufficiently overlapped region of the WAXS data into the SAXS profile. This methodological approach allows for a comprehensive and detailed interpretation of the scattering phenomena, leveraging the unique and complementary attributes of both SAXS and WAXS techniques. During the analysis, we converted the profile into a $4\pi Iq^2$ versus q (Kratky plot) to more clearly features that can be associated with the SBUs solute-rich cluster, mesoscale-ordered structure, or crystallite nanoparticle.

1.5 ZIF-8 nucleation hypothesis

Figure S1 clearly illustrates the mechanism and structural transformations of ZIF-8 nucleation as revealed by molecular dynamics simulations. HmIm ligands to form hydrate complexes, resulting in an alkaline solution (pH = 11.03). Initially, Zn^{2+} forms a hex-coordinated complex with water molecules. This strong Zn^{2+} -water interaction appears to hinder SBUs formation, but the HmIm- N_3 amine site captures a proton from zinc hydration water, disrupting the $\text{Zn}(\text{H}_2\text{O})_6^{2+}$ water cage. (ii) The liberated HmIm ligands can then coordinate to the zinc cavity site to form (iii) the zinc hydronium complex. This complexation lowers the pK_a of $\text{N}_1(\text{H})$ from 14.2 to 10.3, an "acidity flip" that triggers subsequent stages of ZIF-8 nucleation. Enabled by this, HmIm's amine $\text{N}_1(\text{H})$ donates protons, creating a Zn-HmIm hydrate complex network that leads to the formation of SBUs in **Stage I**, decreasing the solution pH to 9.65. Dehydration subsequently drives the development of zinc SBUs networks, transitioning to a mesoscale-ordered structure in **Stage II** and finally to a crystal in **Stage III**

1.6 ZIF-8 deprotonation mechanism simulations

To unveil the intricate growth mechanisms of MOFs, MD simulations were carried out relying on the force field revision and optimization of electrostatic and van der Waals forces to mimic molecular interactions.¹⁻² In this study, the different 12-6 Lennard-Jones (LJ) nonbonded parameters were adopted,³ thereby enriching the portrayal of the interactions between Zn^{2+} ions and their ligands or waters. Our model's accuracy augments the depth of our insights into MOF growth dynamics. Specifically, for the ZIF-8 deprotonation and self-nucleation, the AMBER program,⁴ equipped with a custom-designed force field and supplemented with GAFF parameters.⁵ Our system's initial setup comprised 96 Zn^{2+} and 960 HmIm molecules, prepared using the Packmol package,⁶ in a $60^3 \text{ \AA}^3$ simulation box with a 0.4 g/cm^3 density, ensuring a minimum $4.0 \text{ \AA}$ intermolecular spacing. The aqueous environment was emulated with 960 TIP4P-ew water.⁷ Bond constraints with hydrogen atoms were implemented using the LINCS algorithm, while long-range electrostatic interactions were managed using the Ewald (PME) method. Our simulations employed a $10 \text{ \AA}$ boundary for both types of interactions and used the leapfrog algorithm, setting a 1 fs time-step. After system equilibration at 300 K , and upon achieving peak coordination, the system was compacted by reducing its volume by $0.1 \text{ \%}/\text{ns}$ to a density of 0.5 g/cm^3 . Ensuing this, a sequential deprotonation (25% at density $0.5, 0.6, 0.7$, and 0.85 g/cm^3 , **Figure 2**) of the HmIm was initiated, in line with experimental observations. This scheme continued until a final density of 0.85 g/cm^3 was achieved.

1.7 ZIF-8 nucleation mechanism simulations

For the ZIF-8 nucleation process, the AMBER program,⁴ equipped with a custom-designed force field and supplemented with GAFF parameters (**Table S1** and **Table S2**).⁵ Our system's initial setup comprised 800 Zn^{2+} and 2000 mIm⁻ molecules, prepared using the Packmol package,⁶ in a $140^3 \text{ \AA}^3$ simulation box with a 0.21 g/cm^3 density, ensuring a minimum $4.0 \text{ \AA}$ intermolecular spacing. The aqueous environment was emulated with 8000 TIP4P-ew water.⁷ Bond constraints with hydrogen atoms were implemented using the LINCS algorithm, while long-range electrostatic interactions were managed using the Ewald (PME) method. Our simulations employed a $10 \text{ \AA}$ boundary for both types of interactions and used the leapfrog algorithm, setting a 1 fs time-step. In the initial stage, to ensure an even dispersion of the structures, we set a relatively low density. This allowed us to observe the solute dispersing uniformly in the aqueous solution. After the system reaches equilibrium at 300 K , the density of the system is increased to 0.5 g/cm^3 through a constant pressure system in NPT ensemble. Subsequently, the temperature was increased from 300 K to 1000 K . Then run NVT equilibrated, the structural aggregation became more pronounced, and the emergence of pores was observed with the cavities represented by yellow spheres (**Figure 2**), which were cage-like formations resembling crystal structures.

1.8 DFT calculation for zinc hydration cluster broken in early stage

The initial structure was derived from classical molecular dynamics simulation in which $\text{Zn}(\text{H}_2\text{O})_6^{2+}$ has been formed. Geometry optimization has been carried out by employing the B3LYP⁸ level of theory and cc-pVDZ⁹⁻¹⁰ as the basis set. We have used water as a solvent in the PCM (polarizable continuum

model)¹¹ model. The dynamical stability of the proposed molecules has been investigated by performing Atom Centered Density Matrix Propagation (ADMP) molecular dynamics studies.¹² We generated a 3400 fs trajectory at room temperature with a time step of 1 fs. After structure optimization, Zn²⁺ ions are represented by Zn(H₂O)₆²⁺ (**Figure S2a**). In ADMP-MD results, it is observed that proton transfer (PT) occurs between Zn(H₂O)₆²⁺ and HmIm. The proton of the water molecule is transferred to HmIm, forming both H₂mIm and Zn(OH)⁺(H₂O)_x. Finally, the zinc hydration reaches the 4-coordinates state, resulting in the formation Zn(OH)(H₂O)₃⁺ (**Figure S2b**). In our hypothesis model, we speculate that the environment at this time lets neighboring HmIm have the opportunity to coordinate with Zn²⁺ in the early stage.

1.9 Characterization of ZIF-8 Nucleation at 28 °C

In this study, we investigated the nucleation process of ZIF-8 at various temperatures, including SWAXS data at 10 °C and 28 °C (**Figure S3**). For clearer observation of these processes, we converted the data into Kratky plots (**Figure S4**). The Kratky plots reveal that at 28 °C, the transition from SBUs solute-rich cluster to mesoscale-ordered structure takes about 440 seconds (**Figure S5**), indicating a faster nucleation reaction rate of ZIF-8 at higher temperatures. Additionally, we observed an increase in temperature causes the scattering signals of both SBUs solute-rich cluster and mesoscale-ordered structure to shift to higher q -values. This shift suggests that the structural sizes of these clusters formed at higher temperatures are smaller, as detailed in **Table S3**.

1.10 SWAXS data fitting with atomic structure models

A structure with scattering density $\rho_a(r)$ was surrounded by water with an average scattering density ρ_0 . The hydration shell was approximated by a border layer of effective thickness Δ with density ρ_b that might differ from ρ_0 . The simulated SWAXS intensity (**Figure 3a**) from such particles in dilute solution was proportional to the averaged scattering of a single particle:¹³

$$I(s) = \langle |A_a(s) - \rho_s A_s(s) + \delta\rho_b A_b(s)|^2 \rangle_\Omega \quad (1)$$

$$\rho_b(r) = \begin{cases} 1 & F(\omega) \leq r \leq F(\omega) + \Delta \\ 0 & 0 < r < F(\omega) \text{ or } r > F(\omega) + \Delta \end{cases} \quad (2)$$

where $A_a(s)$ is the scattering intensity from the particle in vacuum, $A_c(s)$ is the scattering intensity from the excluded volume, and $A_b(s)$ is the scattering intensity from the hydration shell, both with unitary density, $\delta\rho = \rho_b - \rho_0$, and $\langle A \rangle_\Omega$ stands for the average overall particle orientations [Ω is the solid angle in reciprocal space, $s = (s, Q)$]. The average atomic radius is 1.623 Å, as calculated by Svergun,¹³ and the concentration of the solvated shell is 0.03 (electron density of the solvent: 0.334 e⁻¹/Å³). According to the spherical harmonic method, we generated the CRY SOL program. The maximum value of the q was 1.2 Å⁻¹, and the number of points was 121. The Fibonacci grid order was 17 and the maximum number of harmonics was 15.

1.11 SBU solute-rich cluster, mesoscale-ordered structure, crystallite nanoparticle and crystallite size determination approach

The size can be determined by converting the SWAXS profile into a Kratky plot and identifying the peak center (q_c).¹⁴ The radius of gyration (R_g) can be obtained through equation (3), assuming the nanoparticle are spherical as equation (4) with a fractal dimension value of 3.96, which is close to 4.¹⁵

$$q_c \times R_g = \sqrt{3} \quad (3)$$

$$R = \sqrt{\frac{5}{3}} \times R_g \quad (4)$$

However, due to the limitations in our data collection range, we could not observe the peak center (q_c) for larger structures, and thus sizes larger than 182 nm are indicated as >182 nm. Given in **Figure 4c**, the size of mesoscale-ordered structure and crystallite nanoparticles can be estimated by identifying the minimum intensity of the scattering vector, q , at the first form factor minimum ($q_{\min 1}$) can numerically be found to be in equation (5)^{14, 16}

$$R = 4.49/q_{\min 1} \quad (5)$$

where R is the radius, and thus $2R$ is the nanoparticle size. Another approach $q_{interval}$ is from sphere form factor interval, and the relationship between radius and $q_{interval}$ can be expressed in equation (6):¹⁴

$$q_{interval} \times R = \pi \quad (6)$$

The Scherrer equation is a formula used to estimate the crystallite size within a material, based on the width of the peak in the WAXS Miller indices.¹⁷ It is expressed in equation (7)

$$D_{hkl} = K\lambda/\beta\cos\theta \quad (7)$$

where D_{hkl} is the mean size of the crystallites, K is the shape factor 0.94 for ZIF-8, λ is the X-ray wavelength, β is the full width at half maximum (FWHM) of the Miller indices peak, and θ is the Bragg angle. We have listed the crystallite nanoparticle and crystallite size determined by different methods or Miller indices (**Table S3**). **Figure 4c** illustrates the size variations over time (per minute) of the crystallite nanoparticle (q_{min1}) measurements and the corresponding crystallite size (D_{011}). It is noticeable that the sizes of both structures tend to stabilize after approximately 2326 seconds.

1.12 SWAXS data fitting with a polydisperse spheres model

Data were calculated form-factor scattering by polydispersed sphere model via SASview. This study analyzes small-angle scattering data to investigate the formation dynamics of clusters and particles. The method utilizes a monodisperse sphere model to represent clusters.^{14, 18} In contrast, particles are modeled as polydisperse spheres governed by a Schulz size distribution $f(R)$, which accounts for size variance within the particle system.

$$f(R) = (z + 1)^{z+1} \left(\frac{R}{R_{av}} \right)^z \frac{e^{-\frac{R}{R_{av}}}}{R_{av}\Gamma(z+1)} \quad (8)$$

$$I(q, R) = scale \times \int_0^\infty f(R) R^6 P(q, R) dR \quad (9)$$

For polydisperse systems, the overall scattering intensity is integral to the size distribution, combining the radius-dependent form factor with the Schulz distribution, which quantifies the fraction of particles of a certain size. This distribution is characterized by its mean radius R_{av} and a polydispersity index p , which relates to the variance of particle sizes in the system (**Figure 4c**).¹⁸

$$z = \left(\frac{1}{p^2} \right) - 1 \quad (10)$$

$$p = \sigma/R_{av} \quad (11)$$

1.13 Kinetic analysis

The equilibrium constant for ZIF-8 formation, eq. (12), has been previously estimated to be $10^{13.4}$,¹⁹

$$K_M = \frac{[Zn(mIm)_2]}{[Zn^{2+}][mIm^-]} \quad (12)$$

Hence, using the pH value, the presumed unreacted concentrations of Zn^{2+} and mIm^- in the solution can be computed, which allows for the deduction of the residual concentration post-reaction and the establishment of the reaction yield. However, outside the equilibrium state, these constants are not applicable for calculating concentration fluctuations. To address this, the ion concentrations at equilibrium were calculated using the pH value once equilibrium was attained. This, in conjunction with the scattering intensities from SWAXS profile at equilibrium, allowed us to estimate concentration variations at different stages of the reaction. Increased temperature leads to three stages of microstructure changes as follows: **Stage I**: The water activity of $HmIm$ increases as the water molecules move faster, promoting the release of OH^- and the collision to break down the structure of the zinc hydrate cluster. It also increases the collision frequency between the zinc hydrate cluster and $HmIm$, accelerating the release of mIm^- protons. The PTs rate increases with increasing temperature, accelerating the formation and nucleation of SBUs. **Stage II**: Increased temperature accelerates collisions between clusters. Because hydrogen bonding is directional and electrostatic Coulomb force is a strong force, these two forces favor the effective phase coupling of SBU networking clusters, promoting the growth rate of mesoscale-ordered

structure growth. **Stage III:** Increased temperature accelerates the generation and transformation of mesoscale-ordered structure growth into crystallite nanoparticle.

The three stages of microstructures characterize the reaction sequence and the change in the limiting reagent Zn^{2+} concentration over time, providing a clear understanding of the microstructure transformation rate at each stage. **Stages I and II** were found to fit the first-order reaction equation and sigmoid equation (orange and purple profiles in **Figure 5a**), with reaction rate constants of 10^{-1} and 10^{-3} per second, respectively; **Stage II** is revealed to be the rate-determining step during the 210-2326 seconds for the nucleation coalesce growth to mesoscale-ordered structure. While the pH changes to 9.69 at the SWAXS capture at $t = 70$ seconds, it reflects the zinc hydration concentration of 1.07×10^{-3} M about 8.5 % remain in aqueous. This implies that within a short time, the PTs driving the zinc water hydration structure have followed the pH drop, transforming primarily into SBUs solute-rich clusters. Thus, the SBUs solute-rich cluster concentration corresponds to 1.16×10^{-2} M (91.4 %). Comparing the decline in SBUs solute-rich cluster intensity to the growth of the mesoscale-ordered structure allows us to quantify the Zn^{2+} portioned to each of the homogeneous solutions, SBUs solute-rich cluster, mesoscale-ordered structure, and crystallites at a specific time. **Table 2** elaborates the corresponding quantities of pivotal structural transformation at a specific stage with time. When the SBUs solute-rich cluster into mesoscale-ordered structure, the sigmoid function provides valuable insights into the dynamics of our system, particularly in terms of the maximum growth rate and the time point. Such information is particularly valuable for predicting future behavior and optimizing experimental conditions.

1.14 Impact of $\text{HmIm}/\text{Zn}^{2+}$ Ratio on Mesoscale Order and ZIF-8 Formation

Figure S6 depicts the synthesis outcomes at a $\text{HmIm}/\text{Zn}^{2+}$ ratio of 19, initially showing the emergence of solute-rich clusters. Notably, after a duration of 1200 seconds, the formation of mesoscale-ordered structures remains absent, suggesting that lower ratios may be unfavorable for ZIF-8 structure development. This observation is likely attributable to the lower HmIm ratio leading to a decreased pH, which impedes the deprotonation of $\text{N}_1(\text{H})$. Consequently, this scenario suggests an increased activation energy for mesoscale-ordered structure formation at reduced ratios. The findings highlight the crucial influence of pH and HmIm concentration on enabling the conditions necessary for ZIF-8 crystallization.

1.15 Pepsin interaction and encapsulation in ZIF-8 nucleation process

After adding protein, changes of mesoscale-ordered structure were observed in the SAXS profile over time (**Figure S7**). For clarity, **Figure S8** shows the SWAXS intensity of the mesoscale-ordered structure and the shift in the q vector during the nucleation of pure ZIF-8 and protein@ZIF-8 at 28 °C. Over time, the SAXS intensity for either ZIF-8 or LYZ@ZIF-8 gradually increases, and the Kratky plot's mesoscale-ordered structure peak center stabilizes at a consistent value. This indicates that lysozyme does not significantly interact with ZIF-8. In contrast, for PEP@ZIF-8, a marked decrease in intensity begins at 750 second, suggesting structural precipitation. Simultaneously, the q vector shifts from $0.0035 \text{ \AA}^{-1}$ back to approximately $0.004 \text{ \AA}^{-1}$, implying that larger ZIF-8 structures are precipitating. However, this precipitation could also be due to the inherent larger size of the structures, which tends to precipitate in aqueous solutions. Compared to the crystal structures at 10 °C, which are larger than those in PEP@ZIF-8, this precipitation is not solely due to size. Therefore, we believe this precipitation results from interactions between ZIF-8 and pepsin, differing from the normal growth pattern. From this, we estimate that approximately 41% of the ZIF-8 crystals are interacting with proteins. In the scenario where we assume Zn^{2+} are the limiting reagent, we can determine the theoretical yield of ZIF-8 by calculating the molar quantity of Zn^{2+} . Additionally, this allows us to ascertain the amount of ZIF-8 that interacts with pepsin.

$$0.25\text{mL} \times 97.5\text{mM} = 2.4375 \times 10^{-5} \text{ mol} \quad (13)$$

$$2.4375 \times 10^{-5} \text{ mol} \times 227.5818 \frac{\text{g}}{\text{mol}} \times 95.7\% \text{ (theoretical yield)} = 5.31 \text{ mg} \quad (14)$$

$$5.31 \text{ mg} \times 41\% = 2.17 \text{ mg} \quad (15)$$

If 2.0 mg of pepsin is completely encapsulated, the encapsulation efficiency would be.

$$\text{protein} \div (\text{protein} + \text{ZIF-8}) = 2.0 \div (2.0 + 5.31) = 27\% \quad (16)$$

Because the encapsulation efficiency is calculated using weight percent concentration, the encapsulation rate inherently tends to be a lower value. However, according to our previous studies,²⁰ the actual encapsulation efficiency is only about 10%. Therefore, the actual amount of pepsin encapsulated is only 0.59 mg, this implies that the remaining 1.41 mg of pepsin merely interacts with ZIF-8 without being encapsulated.

1.16 Temperature-dependent pH variations in ZIF-8 nucleation process

Figure S9 presents the pH variations observed during the nucleation process of ZIF-8 across a range of temperatures. Initially, all samples displayed a pH of approximately 11.03. With the addition of zinc nitrate, the reaction commenced, resulting in a decrease in pH value. Elevated temperatures correspond to a more pronounced decline in pH value. In contrast, at lower temperatures, the pH drop is more moderate, with an initial rise before eventual stabilization and further decrease. This pattern is largely attributable to the different reaction rates induced by temperature variations. The initial decrease in pH is due to the coordination of OH⁻ ions with Zn²⁺ ions. At higher temperatures, mesoscale-ordered structures form rapidly, decreasing the concentration of mIm⁻ in the solution and leading to a continued dissociation of HmIm into mIm⁻ and H⁺, thus causing a steady pH decrease. At lower temperatures, the slower formation of mesoscale structures is likely the cause of the temporary pH increase, before a more gradual decline as the reaction progresses.

1.17 First-order reaction and sigmoid model fitting

The First-order reaction is likely dominant at the early stages of -52-210 seconds during the nucleation process (**Figure 5a**). The exponential decay and growth of species with time reflect that the displacement of water molecules coordinated to Zn²⁺ to form tetra-mIm-coordinated zinc ions as SBUs and the assembly of SBUs into the growth of SBUs solute-rich cluster are the primary focus. The rate expression is given by rate law as a first-order reaction in equation (17).²¹

$$[\text{Zn}] = [\text{Zn}]_0 e^{-kt} \quad (17)$$

the rate constant k can be calculated at a specific temperature, such as 10 °C (283 K), for an initial concentration $[\text{Zn}]_0 = 1.27 \times 10^{-2} \text{ mol L}^{-1}$, and the reactants at HmIm/Zn²⁺ molar ratios of 57 across 3600 seconds, thus $k = 0.15 - 1.36 \text{ s}^{-1}$. Herein, Eyring equation is that it allows us to estimate the activation energy 16.67 - 18.15 kcal/mol for the early stage of PTs dominance the formation of SBUs precursors and predict the rate at a temperature.²¹⁻²² The sigmoid function is shown in equation (18):

$$y = a / (1 + e^{-k(t-t_c)}) \quad (18)$$

y is the Zn²⁺ concentration or scattering intensity, a is the maximum value that y can reach (carrying capacity), k is the growth rate, t is the reaction time, and t_c is the time of maximum growth (**Table S4**). The Eyring equation describes the relationship between temperature and the rate constant of a reaction as equation (19):²²

$$\ln\left(\frac{k}{T}\right) = -\frac{\Delta H^\ddagger}{R} \times \frac{1}{T} + \ln\left(\frac{\kappa k_B}{h}\right) + \frac{\Delta S^\ddagger}{R} \quad (19)$$

κ is transmission coefficient and assumed for 1; k is rate constant; k_B is Boltzmann's constant; T is the absolute temperature; h is the Planck's constant; $\Delta G^\ddagger$ represents the Gibbs free energy change of the transition state; R is the universal gas constant. The Gibbs free energy change of the transition state, $\Delta G^\ddagger$, can be calculated in equation (20):²¹

$$\Delta G^\ddagger = \Delta H^\ddagger - T\Delta S^\ddagger \quad (20)$$

1.18 SWAXS background subtraction

In the processing of scattering data, it's necessary to subtract the background (HmIm and CTAB solution) signal from the measured sample signal. This procedure involves measuring the X-ray

Transmission (TM) of both the sample and the background to standardize the scattering intensity. **Figure S10** highlights the challenges we encountered during the TM correction process. Taking data measured at 10 °C as an example, the initial TM value for the background was recorded as 0.756. However, once the ZIF-8 nucleation reaction commenced, it was imperative to measure the SWAXS data immediately. This urgency arises due to the two-minute transition time required to switch between measuring SWAXS and TM, preventing simultaneous TM measurements. Consequently, the TM measurement for the sample could only be conducted after completing the SWAXS data collection, revealing a TM value of 0.698. The TM value during the intermediate stages is therefore reasonably assumed to be between 0.698 and 0.756. Although this influences the precision of the background subtraction, the overall effect on the signal is not substantial.

2. Figure Section

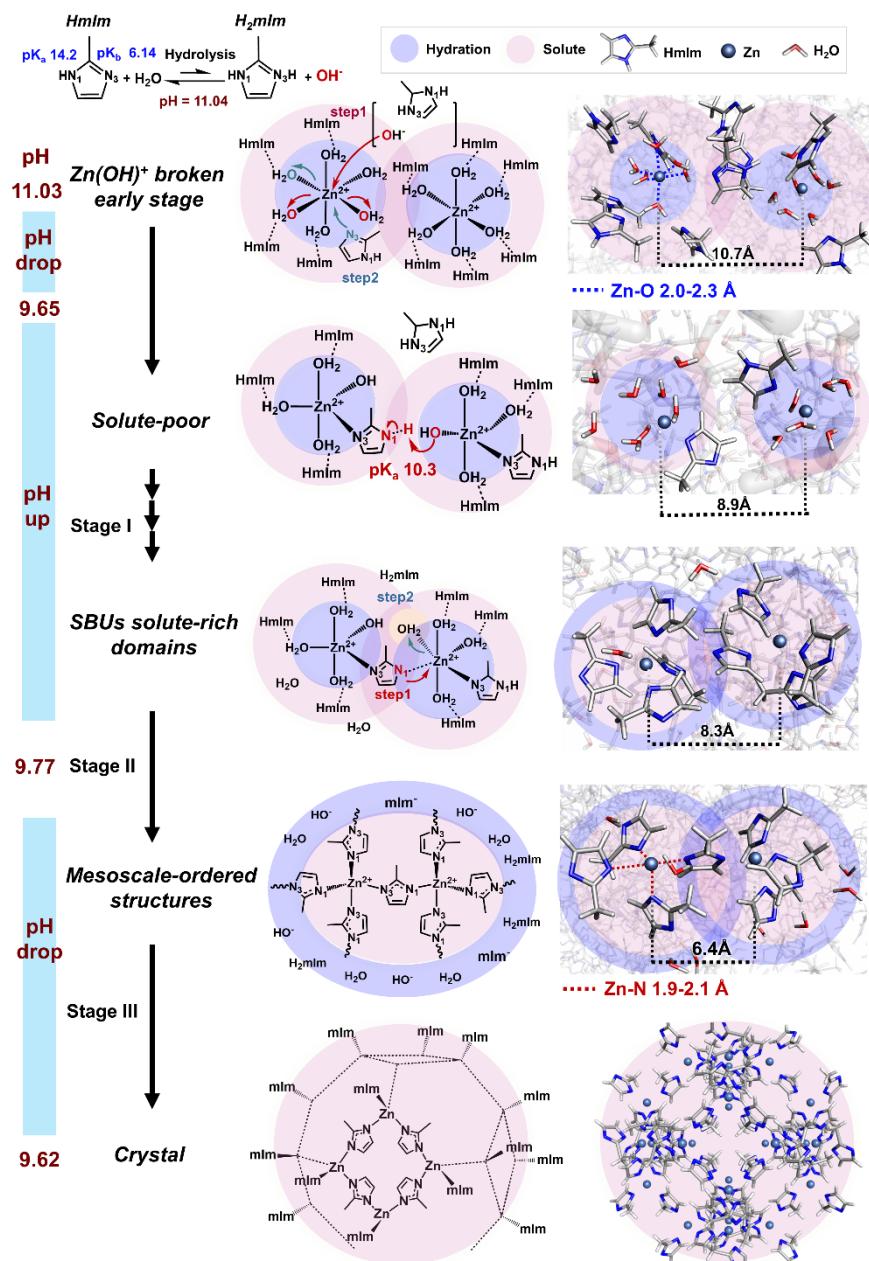


Figure S1. The hypothesis of proton transfer (PT) mechanism for the ZIF-8 nucleation process involves three stages.

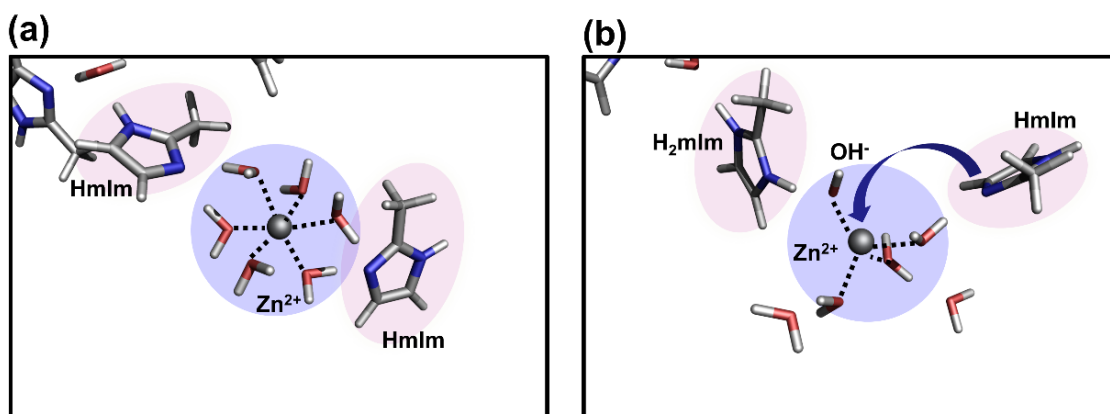


Figure S2. Coordination structure of Zn^{2+} derived from ADMP-MD simulations. (a) The Zn^{2+} is initially surrounded by six water forming the $\text{Zn}(\text{H}_2\text{O})_6^{2+}$ complex, and some HmIm present in proximity. (b) ADMP-MD calculations illustrate the formation of a $\text{Zn}(\text{OH})(\text{H}_2\text{O})_3^+$ complex, where the coordination by OH^- leads to a change in the coordination number and the geometry of the complex, consequently allowing HmIm molecules the opportunity for coordination with Zn^{2+} .

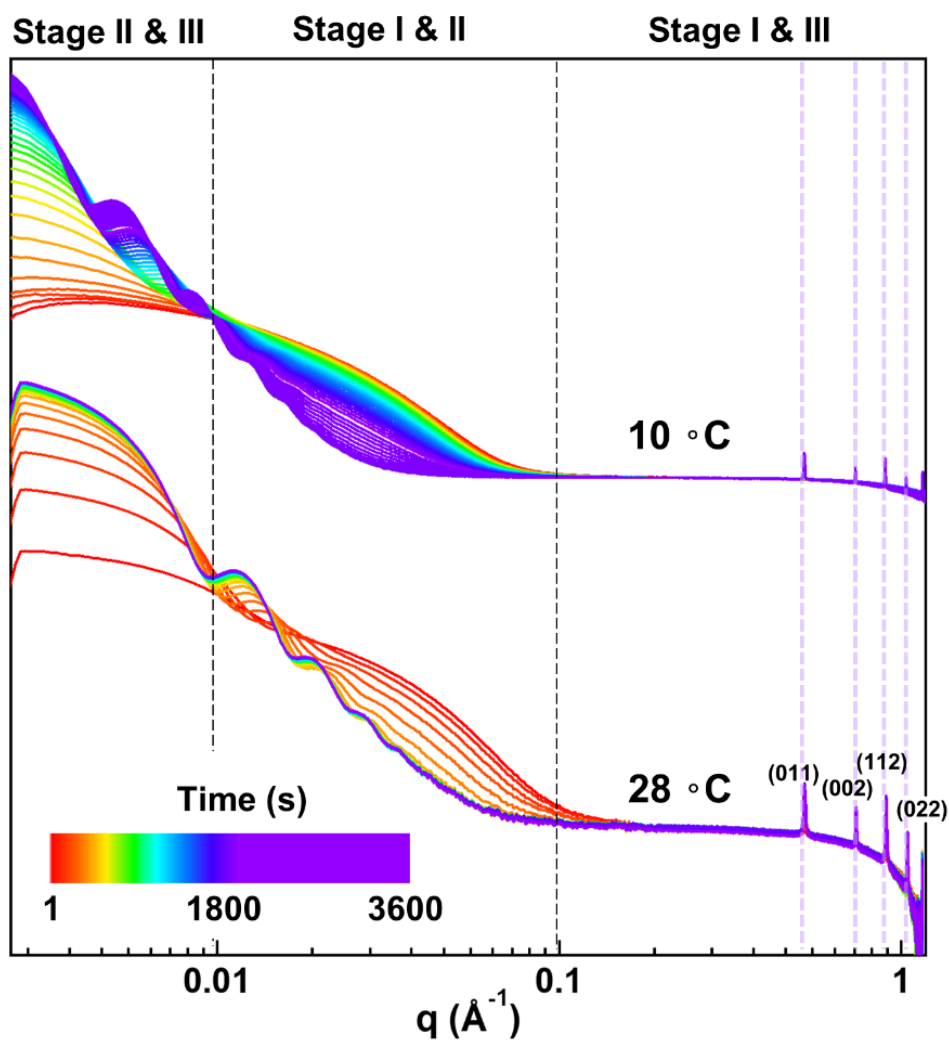


Figure S3. Time-resolved SWAXS profile of ZIF-8 formation at 10 °C and 28 °C. This figure displays the SWAXS profile over a period of 3600 seconds, capturing the evolution of the ZIF-8 precursor intermediates at HmIm/ Zn^{2+} molar ratio of 57. The color gradient represents elapsed time in minutes to enhance clarity, distinctive features in the profile correlate with the stages of nucleation and growth at each temperature.

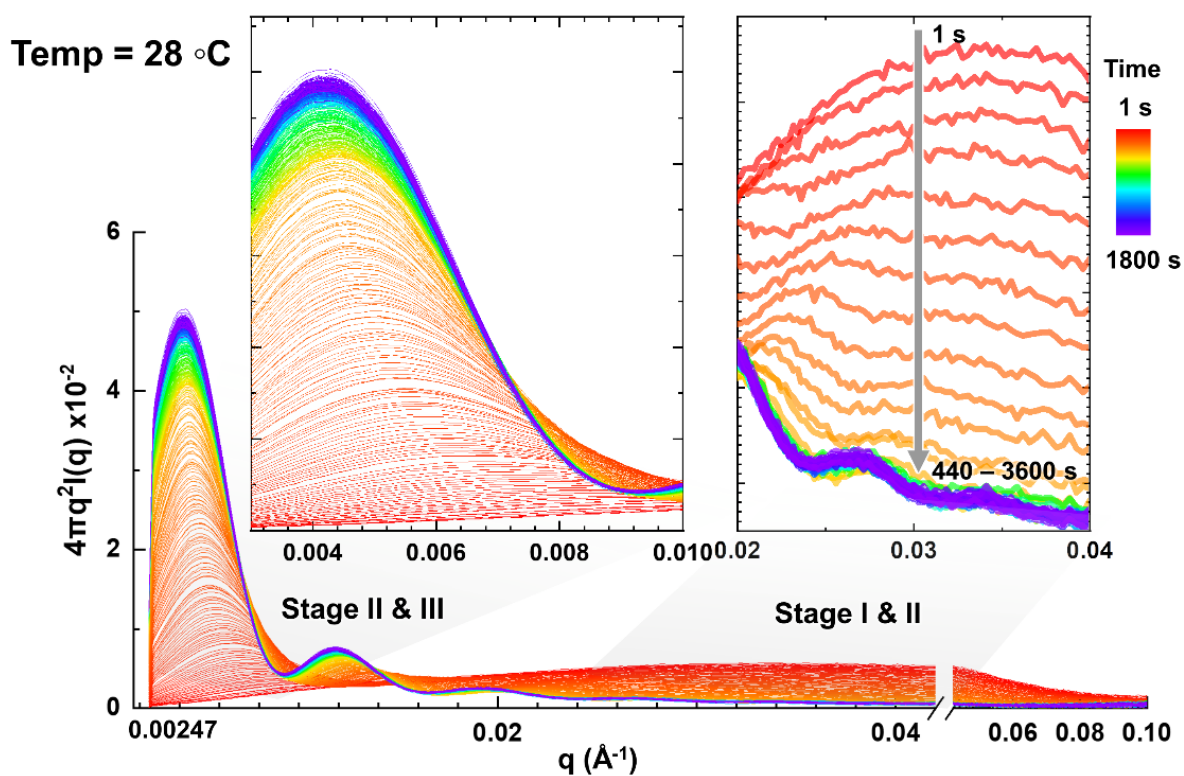


Figure S4. Kratky plot analysis of ZIF-8 nucleation at 28 °C. The plot captures the transformation during Stage II of the nucleation process, where the system evolves from SBUs solute-rich cluster to mesoscale-ordered structure. The transformation is observed around 1 - 440 seconds with the development of 75-85 nm mesoscale-ordered structure from 13 nm SBUs solute-rich cluster.

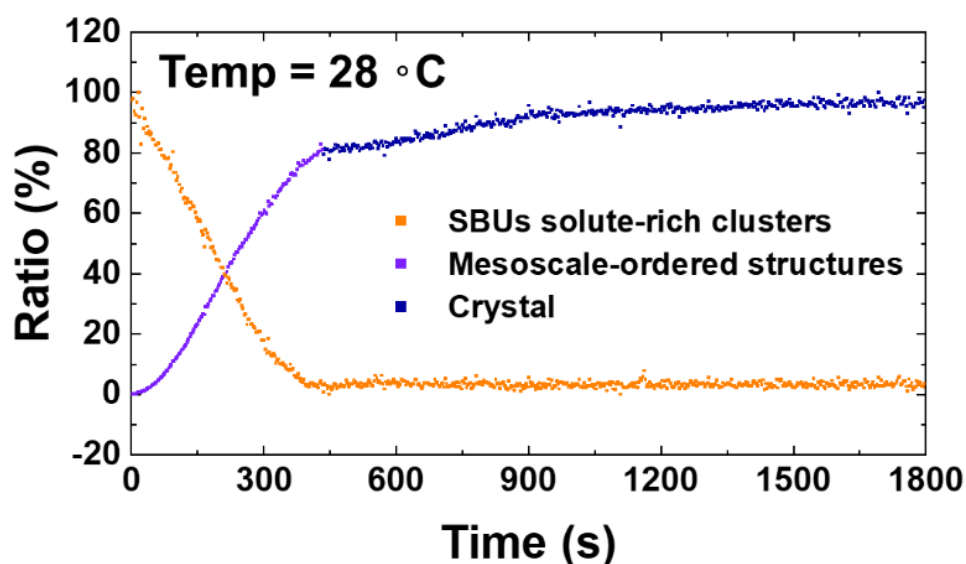


Figure S5. Temporal quantification of the SBUs solute-rich cluster and mesoscale-ordered structure at 28 °C. The graph depicts the proportional shift from SBUs solute-rich cluster to mesoscale-ordered structure during ZIF-8 nucleation. A swift transformation is evident within 440 seconds, where SBUs solute-rich clusters diminish as mesoscale-ordered structure concomitantly increase, reflecting the nucleation's temperature-sensitive kinetics. This marked conversion demonstrates the enhanced assembly rate at 28 °C, suggesting temperature as a critical factor in the nucleation and growth of ZIF-8 structures.

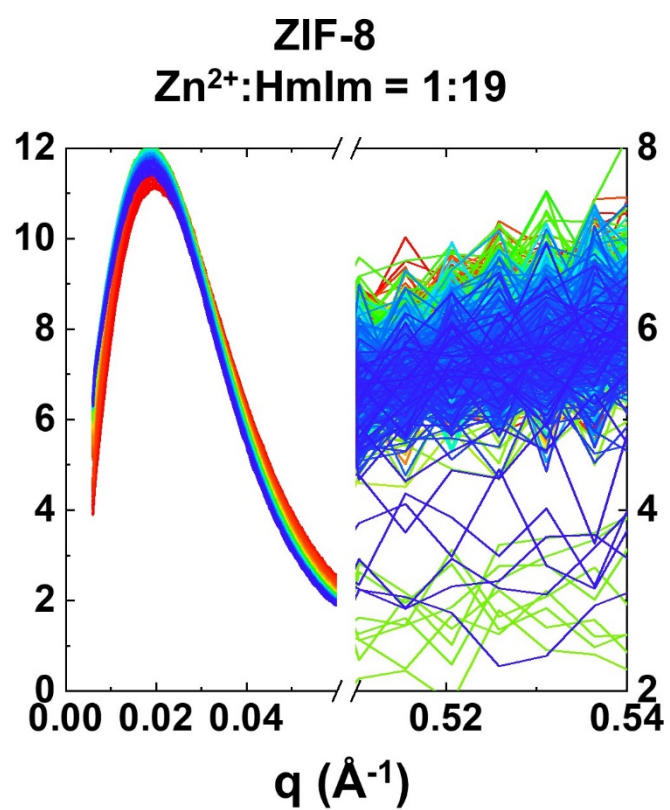


Figure S6. Time-dependent changes in the SWAXS Kratky plot during the reaction process of ZIF-8 19 HmIm/Zn²⁺ ratio.

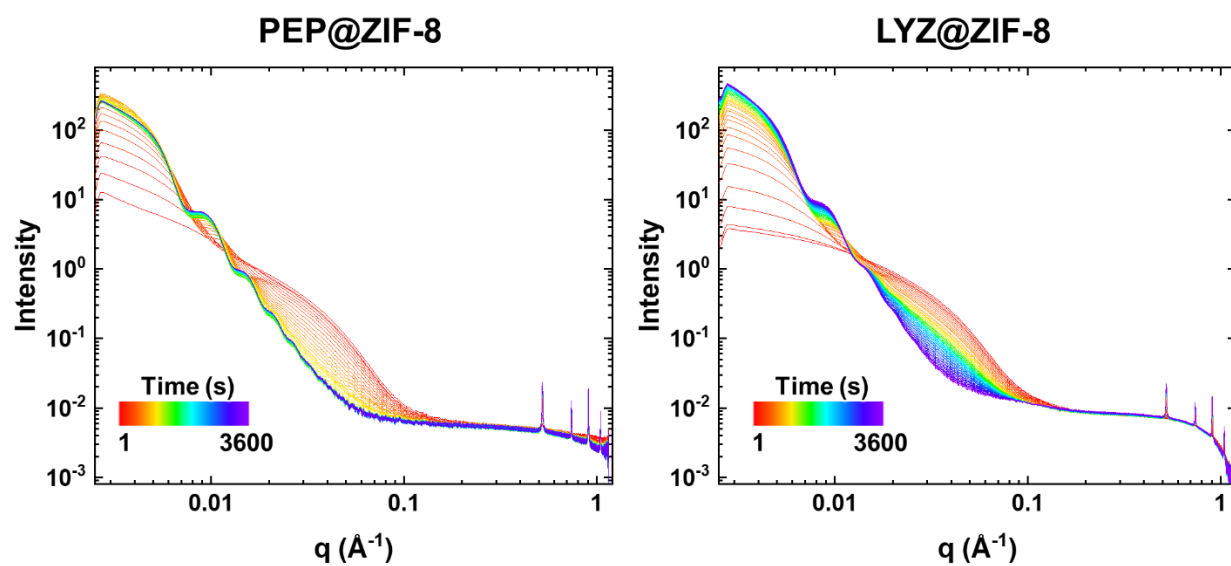


Figure S7. Time-resolved SWAXS profile of ZIF-8 formation with pepsin@ZIF-8 or lysozyme@ZIF-8 at 28 °C.

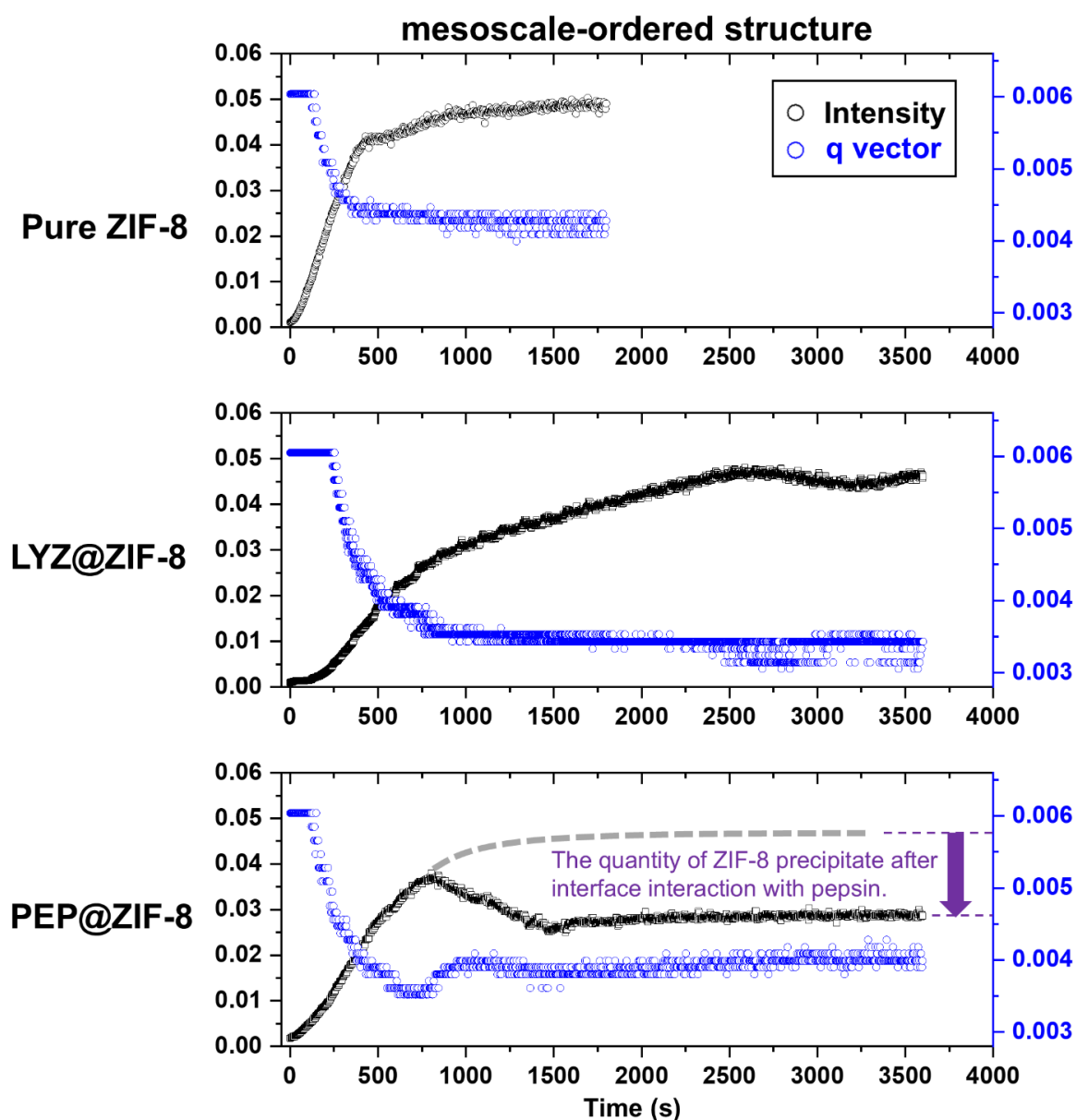


Figure S8. Temporal quantification of the mesoscale-ordered structure at 28 °C. The graph depicts the mesoscale-ordered structure SAXS intensity and q vector shift during ZIF-8 nucleation in different system. It is clearly observed that there is a significant decrease in structural intensity around 750 seconds, attributed to the ZIF-8 precipitate interacting with proteins. By predicting the original intensity, we can estimate that the interaction ratio between ZIF-8 and pepsin is approximately 41%.

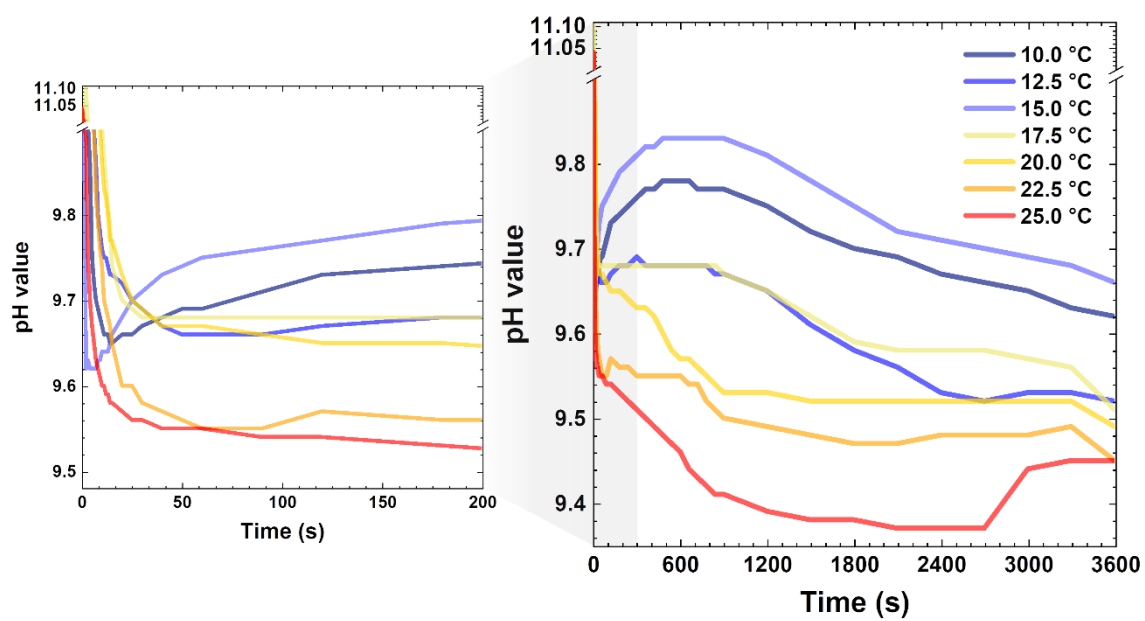


Figure S9. Time-dependent changes in the pH values during the reaction growth process of ZIF-8 synthesis at a 1:57 ratio at different temperatures.

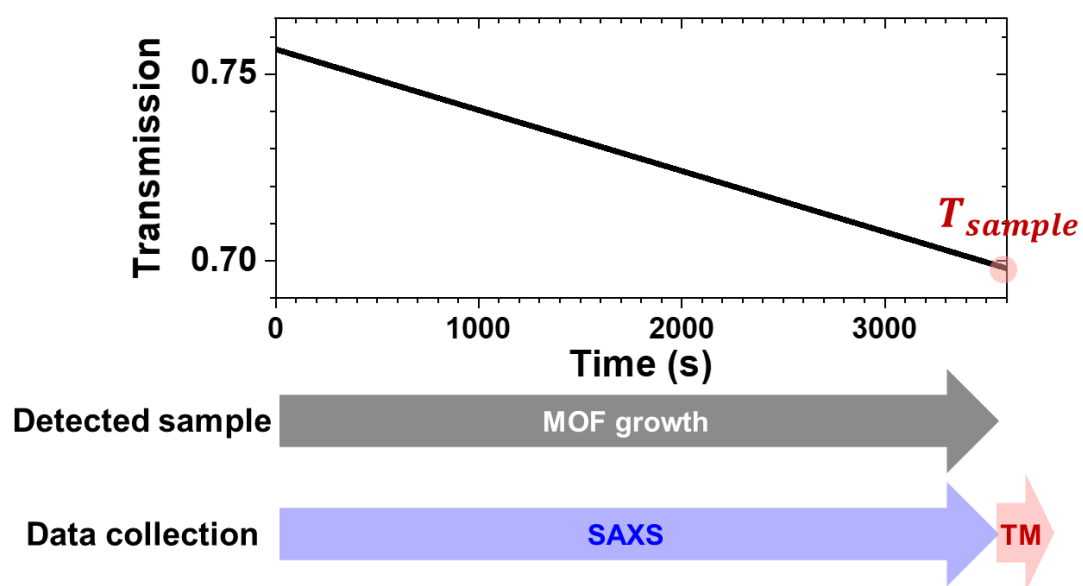


Figure S10. The transmission of the sample changes with the detection time.

3. Table Section

Table S1. Force field between ZIF-8 and HmIm

Atom 1	Atom 2	A coefficient	B coefficient	R_{ij}	ϵ_{ij}
Zn	Zn	2135.414220	4.630038	3.120000	0.002510
Zn	cc,cd	44464.334100	51.117223	3.468000	0.014691
Zn	nb	35725.453700	47.580130	3.384000	0.015842
Zn	c3	50150.033500	57.653635	3.468000	0.016570
Zn	h4	2878.601200	8.405244	2.969000	0.006136
Zn	hc	4020.062250	10.046807	3.047000	0.006277
Zn	OW	38386.517100	55.706561	3.335931	0.020210
Zn	HW	0.000000	0.000000	0.000000	0.000000
Zn	EP	0.000000	0.000000	0.000000	0.000000

Table S2. Modification of Zn atomic force field parameters.

Atom type parameter	Zn		nc	
	r	ϵ	r	ϵ
Amber Force Field	1.252	0.00250973	1.824	0.1700
New Force Field	1.6276	0.00250973	1.824	0.1000

Table S3. Comparative analysis of semi-crystalline nanoparticle and crystallite sizes at various temperatures derived from SWAXS in different q-regions.

T (°C)	Crystallite nanoparticle size (nm) ^a			Crystallite size (nm) ^b			
	<i>q_{min1}</i>	<i>q_{interval}</i>	<i>q_c</i>	D₀₁₁	D₀₀₂	D₁₁₂	D₀₂₂
28	97	85	105	80	79	75	82
25	116	103	124	89	92	80	89
22.5	225	197	>182	120	126	111	113
20	197	183	>182	119	121	109	111
17.5	197	183	>182	106	114	98	98
15	337	339	>182	139	141	128	128
12.5	215	357	>182	136	142	128	127
10	214	194	>182	120	126	112	112

^aCrystalline nanoparticle size can be estimated with the SAXS form factor analysis. ^bCrystallite size based on the width of the peak in the WAXS Miller indices.

Table S4. Parameters of Sigmoidal fitting for ZIF-8 nucleation at various temperatures.

T (°C)	[MOS] _{max} (M)	k (1/s)	t _c (s)
28	1.13 x 10 ⁻²	1.04 x 10 ⁻²	253
25	1.16 x 10 ⁻²	8.65 x 10 ⁻³	448
22.5	1.16 x 10 ⁻²	6.49 x 10 ⁻³	381
20	1.17 x 10 ⁻²	5.81 x 10 ⁻³	508
17.5	1.17 x 10 ⁻²	3.27 x 10 ⁻³	984
15	1.17 x 10 ⁻²	3.16 x 10 ⁻³	797
12.5	1.20 x 10 ⁻²	2.75 x 10 ⁻³	1310
10	1.17 x 10 ⁻²	2.26 x 10 ⁻³	1398

^aThrough the sigmoid model $[\text{MOS}]_t = [\text{MOS}]_{\text{max}} / (1 + e^{k(t-t_c)})$ values of $[\text{MOS}]_{\text{max}}$ (maximum mesoscale-ordered structure concentration), k (maximum growth rate), and t_c (half-concentration time) are determined.

4. References

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